

Quiz — 1.5.2 Moral and Ethical Issues — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Answer key

Section A (1 mark each = 8)

1. **B** — Machines doing tasks once done by humans is the automation/workforce issue.
2. **B** — An algorithm reproducing bias from historical data is AI ethics / algorithmic bias.
3. **C** — E-waste and data-centre energy use are environmental drawbacks (A, B, D are benefits).
4. **A** — Blocking access to content is censorship/control of the internet.
5. **B** — This is the definition of the digital divide.
6. **C** — Moving work abroad to cut costs is offshoring.
7. **B** — Monopolies tend to mean higher prices and lock-in (the others are advantages).
8. **B** — The "trolley problem" for autonomous vehicles is about automated decision making and liability.

Section B (8)

9. **[2]** Benefit (1): fast, consistent, handles huge volumes of data without fatigue/no human error from tiredness. Drawback (1): lacks empathy/human judgement, hard to appeal a decision, or unclear who is liable for a wrong/harmful decision.
10. **[2]** AI learns patterns from its **training data** (1); if that data reflects historical or societal **bias** (or is unrepresentative), the model **reproduces or amplifies** that bias, and a "black box" model makes it hard to detect or explain (1).
11. **[2]** Any two, 1 each: cannot access online-only services (banking, government forms); disadvantaged in education/jobs that require digital skills; excluded from cheaper online prices/deals; isolated socially; particularly affects poor, rural, elderly or disabled people — widening inequality.
12. **[2]** Benefit (1): helps prevent/detect crime or terrorism, supports public health, enables personalised services. Drawback (1): loss of privacy, profiling, risk of data breaches/misuse, or a "chilling effect" on free expression.

Section C — indicative content (9)

This is a levels-of-response "**discuss**" question; mark the **whole answer**, not point-by-point. Expect a balanced discussion across multiple stakeholders with a justified conclusion.

Benefits (FOR): - *Supermarket/business*: lower wage costs, faster throughput, fewer queues, more competitive prices. - *Customers*: quicker, sometimes 24/7 service; potentially lower prices. - *Some new roles*: technicians to maintain/repair the systems.

Drawbacks (AGAINST): - *Employees*: job losses / layoffs, especially low-skilled checkout staff; deskilling; regional unemployment. - *Society*: loss of human contact (a problem for elderly/vulnerable shoppers); widening of the digital divide for those uncomfortable with the technology; data/privacy concerns with "just

walk out" tracking and surveillance. - *Ethical*: is it right to prioritise profit over jobs; corporate responsibility to retrain staff.

Stakeholders to name (aim 3–4+): the business/shareholders · employees · customers (incl. elderly/vulnerable) · wider society/local community.

Justified conclusion: must take a clear position, ideally conditional — e.g. *"automation is justified provided staff are retrained or redeployed and some staffed tills remain for those who need assistance."*

Levels of response (9 marks): - **Level 1 (1–3)**: Fragmented, one-sided list (e.g. only job losses or only cost savings). Little development, weak/no use of terminology, no conclusion. - **Level 2 (4–6)**: Some discussion of both sides but mostly descriptive or one-sided; limited stakeholders; conclusion weak, unsupported or absent. - **Level 3 (7–9)**: Balanced and well-developed, several stakeholders, **both** benefits and drawbacks with accurate terminology and sustained reasoning, leading to a **justified conclusion** that takes a reasoned (often conditional) position.

Total: 8 + 8 + 9 = 25